

DJ Statement

Imagine yourself immersed in a crowd - you're front row to your favorite show, it's hot, boiling by the minute but this invaluable moment makes anything bearable. The show begins, fog on the gravel stage sends chills as it reaches your skin. The lights hit, creating a synergistic feeling amongst the crowd that's not spoken about, yet you can feel the emotions it illuminates hidden beneath its scenery - this is my discourse. Theoretical Computer Science, specifically Compilers/Programming Language Theory (PLT) are like a Light Director during a concert - you know it's there but you don't know the intricacies on how it works. You might think the importance of a Light Director is minimal, however, imagine yourself at a concert without lights - how would that experience feel? I would imagine dull and dark - the same applies in a compiler, without its presence, knowing where to go or even what's next is ambiguous. However, the beauty lies in its purpose - just as a Light Designer, a compilers' purpose is taking control. Control of the user's emotions and understanding, allowing them to connect with their environment by a feeling and ensuring they are focused on the particulars: the stage, singer, drummer, guitarists, the program.

This soundtrack is designed to explore said discourse and reflect my journey so far within the discourse community. Each track represents a different perspective, subjectively shaped by the experiences I've had and share insight on why I've become so fascinated with computation, abstraction, and modern technology as a whole.

Virtual Insanity - Jamiroquai (greatest hits)

Jamiroquai is a group known for their unique clothing, vocals, slapping bass-lines, and 'acid' funk style of music. Virtual insanity correlates in particular with my discourse from the recent hype around artificial intelligence. During the song, the quote: "Futures made of virtual insanity, now always seem to be governed by this love we have for useless twisting of our new technology" hits the surrealism of modern ideology with respect to AI which revolves around questioning the ethics of AI and its application. However, I also connect this idea to the growing disconnect between society and the underlying systems that make modern computation possible. Within PLT/Compilers, there is strong emphasis on formal reasoning, determinism, and verification - even when modern technology is treated like 'magic' by the general public. The 'virtual insanity' I associate with my discourse community, revolves around the assumption that increasingly abstract systems can exist without understanding the mathematical and logical foundations beneath them. This idea can lead to misunderstanding, false confidence, and in my opinion a form of 'virtual insanity'.

Duality - Slipknot (hit single)

There is a unique distinction that philosopher Rene Descartes with respect to duality claims and that is, we perceive things in two forms: things in the physical world and things that are mental only to ourselves - "live in our thoughts". This is the same connection that I feel when making programs - that is to say, it feels sort of dualistic in a way when coming up with an idea then proceeding to make it physically for the world. It makes me wonder how many fantastic ideas have been thought of but never made or "shown" for others to appreciate and expand upon.

Pneuma - TOOL (wildcard)

I would not describe Tool as a “band” or a “group”, Tool is an experience and for those who have not had the pleasure of hearing such a unique sound, I will try my best to elaborate why I’ve selected them. The band is backed by the LEGENDARY drummer, Danny Carey - iconically known for his unorthodox style of playing. His ability to express emotion and feeling each snap of the stick sends an exhilarating feeling down your spine. *Pneuma* is Danny’s “magnum opus” - it is a masterpiece, the song begins with a three minute intro which straps you in for the ride, before evolving into a mesh of polyrhythms, tempo changes, and layered instrumentation that feels ambiguous and unpredictable. This unorthodox style is what connects the song to my discourse community within PLT/compiler design. Compilers are full of depth, that is to say - there are many “paths” one could take with respect to being a compiler engineer. One could spend years studying how to build parsers, improve lexical analysis, while another focuses on optimization and formal verification without ever seeing each other's sides. The balance between unpredictability and rigor is what makes both *Pneuma* and PLT so captivating to me.

Gansta's Paradise - Coolio (wildcard)

“I’m so unimportant to the world, yet I’m so important to the world.” This is my personal perspective for Compilers/PLT - why and what do I even mean? How I like to picture it is simple, most take for granted how we have amazing modern technology: cars, phones, internet, yet most technical people do not realize what those things reduce to. Well I’ll tell you, it reduces to a compiler and a programming language, it reduces to my discourse and THAT is “Gansta”.

With respect to the song, *Gangsta's Paradise* refers to entrapment, living in a dangerous and exposed place. Obviously saying there is a relation is hard to justify because there are different stakes, comparing living in danger vs working on a compiler, but the analogy feels similar. For lack of better words, we use the term “blowing up” to express a Compiler failing, or something programmatically wrong. How is this dangerous? Well it’s detrimental that your compiler is consistent and deterministic - without determinism how would we know when to stop or even where to stop? If a compiler is inconsistent, what does that say about every program that relies on compilers or interpreters (which is EVERY program)?